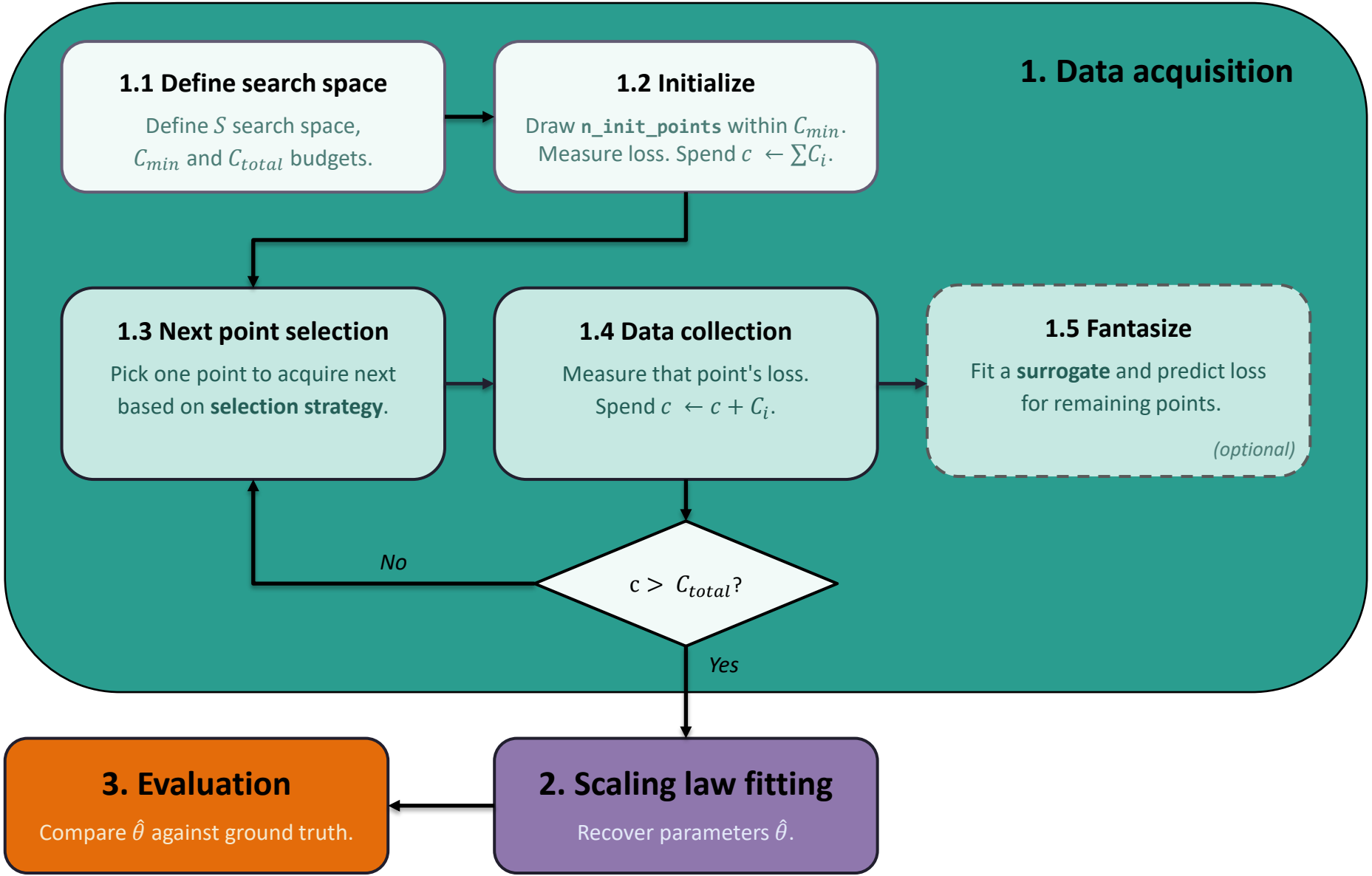

1. Data acquisition
1.1 Define search space
Define $S$ search space, $C_{min}$ and $C_{total}$ budgets.
1.2 Initialize
Draw n_init_points within $C_{min}$. Measure loss. Spend $c \leftarrow \sum C_i$.
1.3 Next point selection
Pick one point to acquire next based on selection strategy.
1.4 Data collection
Measure that point's loss. Spend $c \leftarrow c + C_i$.
1.5 Fantasize
Fit a surrogate and predict loss for remaining points.
(optional)
No
$c > C_{total}$?
Yes
2. Scaling law fitting
Recover parameters $\hat{\theta}$.
3. Evaluation
Compare $\hat{\theta}$ against ground truth.